# Ing. Márk Bartalos

*Full-Stack Software Engineer*

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## ABOUT ME

Full-Stack Software Engineer with 3+ years of professional experience and 5+ years of hands-on software development experience. Experienced in developing web applications across the full stack, with academic and project experience in AI, large-scale data processing, and blockchain technologies. Specialized in React and Next.js, with expertise in blockchain development (Ethereum, Solana, Polkadot) and back-end experience in .NET, Node.js and Rust.

## SKILLS

**Frontend**
React, Next.js, Angular, JavaScript/TypeScript, HTML, CSS, Tailwind CSS, shadcn, Playwright, Vue.js, Nuxt

**Backend**
.NET C#, Node.js, Rust, REST API, Entity Framework, Prisma, GraphQL

**Blockchain**
Solidity, Viem, Polkadot, Ethereum, Solana, Account Abstraction, ZK Proofs

**Databases**
MS SQL, PostgreSQL, MongoDB

**DevOps**
Docker, Kubernetes, Git

**Tools**
Apache Lucene, Apache Spark

## PROFESSIONAL EXPERIENCE

**2Ring,** *Full-Stack Developer* 06/2022 – 09/2024 | Bratislava, Slovakia
*Technologies: .NET, Entity Framework, MS SQL, Rust, Aurelia, Vue.js, Kubernetes*
- Worked on and maintained enterprise contact center dashboards serving 62+ countries including Fortune 500 companies
- Enhanced real-time monitoring applications written using Vue.js and Aurelia framework
- Optimized back-end features in .NET and Entity Framework
- Designed RESTful API services for real-time data visualization and reporting
- Contributed to rewriting data processing modules in Rust for improved performance

**Faculty of Informatics and Information Technologies STU** 09/2024 – 05/2025 | Bratislava, Slovakia
*Full-Stack Developer & Scrum Master*
*Technologies: .NET, IdentityServer, Entity Framework, Angular, RxJS, Angular Material*
- Led team development of HIPA, a medical microscopy platform for Comenius University Faculty of Medicine
- Developed an advanced admin panel for teachers to enable learning content management
- Implemented Angular front-end with Material Components and RxJS for responsive interface

**Dorps (Contract),** *Full-Stack Developer* 03/2025 – 09/2026 | Bratislava, Slovakia
*Technologies: .NET, Angular, Next.js*
- Developed and extended web applications.

- Implemented and maintained RESTful APIs.

**Decenzio (Freelance),** *Front-End/Blockchain Developer* 11/2025 – 06/2026
*Technologies: React, Playwright, Chopsticks (polkadot local testnet), ThreeJS*
- Implemented e2e tests for testing XCM asset transfers locally
- Refactored rendering pipeline on front-end to correctly render images in a 3D space and improved performance using memoization
- Developed a CLI tool for complex starter project scaffolding (multiple frameworks, clients and features)

**Faculty of Informatics, STU - DEUSS Project** 01/2026 – 06/2026 | Bratislava
*Front-End/Blockchain Developer*
*Technologies: React, TanStack Router, Viem, Ethereum*
- Implemented a UI for on-chain bond transfers with extra KYC validation
- Implemented visualization and management functionality for on-chain orderbook
- Migrated the frontend between major contract/indexer releases

## PROJECTS

**ReAnimate,** *AI-powered PDF to ebook conversion platform* 06/2025
Developed a web application for smart conversion of PDF documents to modern ebook and markdown formats using advanced AI OCR.

*Technologies: Next.js, Rust, Prisma, PostgreSQL, Tailwind CSS, shadcn, NextAuth*

- Built intelligent processing pipeline handling diverse PDF structures while preserving document fidelity
- Implemented asynchronous job queue system with real-time progress tracking for concurrent file conversions
- Developed high-performance Rust backend for optimal processing speed and efficiency
- Designed responsive UI using Tailwind CSS and shadcn/ui component library
- Integrated OAuth authentication (Google, GitHub) via NextAuth for secure third-party login

## EDUCATION

**Master's Degree - Computer Science** 09/2024 – 06/2026 | Bratislava, Slovakia
*Faculty of Informatics and Information Technologies STU*

**Bachelor's Degree - Computer Science** 09/2020 – 05/2024 | Bratislava, Slovakia
*Faculty of Informatics and Information Technologies STU*

## PUBLICATIONS

**Building a decentralized application on Solana** 2026
*Faculty of Informatics and Information Technologies STU*
Implemented a gaming-themed NFT marketplace deployed on both Solana and Ethereum, comparing architectural and developer experience differences in decentralized application development across the two chains.

**Practical usage of Zero-knowledge in different use-cases in blockchains** 2024

*Faculty of Informatics and Information Technologies STU*

Implemented a modern Ethereum wallet with passkey support, leveraging zero-knowledge proofs and the ERC-4337 account abstraction standard to explore practical ZK applications in blockchain identity and authentication.

## LANGUAGES

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| English | ●●●●● | Hungarian | ●●●●● |
| Slovak | ●●●●○ | German | ●●○○○ |